

CMB, CMB-SP Booster PM2

Supplement to Quick Guide



English (GB) Installation and operating instructions

Supplement to quick guide.

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Warning
Prior to installation, study the quick guide carefully. Installation and operation must comply with local regulations and accepted codes of good practice.



Warning
The use of this product requires experience with and knowledge of the product.
Persons with reduced physical, sensory or mental capabilities must not use this product, unless they are under supervision or have been instructed in the use of the product by a person responsible for their safety.
Children must not use or play with this product.

1. Symbols used in this document



Warning
If these safety instructions are not observed, it may result in personal injury!



Warning
If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.



Caution
If these safety instructions are not observed, it may result in malfunction or damage to the equipment!



Note
Notes or instructions that make the job easier and ensure safe operation.

2. System sizing



Warning
The system in which the CM Booster is incorporated must be designed for the maximum pump pressure.



The default setting will cause the pump not to stop until it reaches its maximum pressure.

3. Mechanical installation



Warning
Before starting any work on the product make sure that the power supply has been switched off and that it cannot be accidentally switched on.

4. Electrical connection

The electrical connection should be carried out by an authorized electrician in accordance with local regulations.



Warning

Check that the supply voltage and frequency match the voltage and frequency on the nameplate.

Warning

Before making any connections in the unit, make sure that the power supply has been switched off and that it cannot be accidentally switched on.



The unit must be connected to an external mains switch with a contact gap of at least 3 mm in all poles.

As a precaution, the unit must be connected to a socket with earth connection.

We recommend to fit the permanent installation with an earth leakage circuit breaker (ELCB) with a tripping current < 30 mA.

5. Startup

Caution

Do not start the pump until it has been filled with liquid.

6. Alarms

Caution

If a dry-running alarm has been activated, find the cause before the pump is restarted in order to prevent damage to the pump.

6.1 Anti-cycling

Note

In case of a very small consumption, the anti-cycling function may detect this as cycling and stop the pump inadvertently. If this occurs, the function can be disabled.

Disabling the function

To disable the function, set DIP switch 7 to "OFF". See Quick Guide.

6.2 Auto reset

Caution

Set auto reset to "on" only if there is positive inlet pressure.

7. Fault finding chart



Warning

Before starting work on the pump, make sure that the power supply to the pump has been switched off and that it cannot be accidentally switched on.

Fault	Cause	Remedy
1. The green light field for "0 bar" is off even if the power supply has been switched on.	a) The fuses in the electrical installation are burnt.	Replace the fuses. If the new fuses are also burnt, check the electrical installation.
	b) The earth leakage circuit breaker or the voltage-operated circuit breaker has tripped.	Cut in the circuit breaker.
	c) No power supply.	Contact the power supply authorities.
	d) The PM unit is defective.	Repair or replace the PM unit.
2. The green "Pump on" indicator light is on, but the pump does not start.	a) The power supply to the pump is disconnected after the PM unit.	Check the plug and cable connections, and check if the built-in circuit breaker of the pump is switched off.
	b) The motor protection of the pump has tripped due to overload.	Check if the motor/pump is blocked.
	c) The pump is defective.	Repair or replace the pump.
	d) The PM unit is defective.	Repair or replace the PM unit.
3. The pump does not start when water is consumed. The "Pump on" indicator light is off.	a) Too big difference in height between the PM unit and the tapping point.	Adjust the installation, or increase the start pressure. See quick guide.
	b) The PM unit is defective.	Repair or replace the PM unit.
4. System without pressure tank: Frequent starts/stops.	a) DIP switch 5 set to "ON".	Set DIP switch 5 to "OFF". See quick guide
	b) Leakage in the pipework.	Check and repair the pipework.
	c) The non-return valve is stuck in open position.	Clean or replace the non-return valve.
5. System with pressure tank: Frequent starts/stops.	a) The pressure tank has no precharge pressure, or the tank size is insufficient.	Check the tank precharge pressure, and recharge the tank, if necessary. If the size of the pressure tank is insufficient, set DIP switch 5 to "OFF", or replace the pressure tank.
	b) Leaky non-return valve.	Clean or replace the non-return valve.
6. The pump does not stop.	a) The pump cannot deliver the necessary discharge pressure.	Replace the pump.
	b) The start pressure is set too high.	Decrease the start pressure.
	c) The PM unit is defective.	Repair or replace the PM unit.
	d) The non-return valve is stuck in open position.	Clean or replace the non-return valve.

Fault	Cause	Remedy
7. The red "Alarm" indicator light is permanently on.	a) Dry running. The pump needs water.	Check the pipework.
	b) The power supply to the pump is disconnected after the PM unit.	Check the plug and cable connections, and check if the built-in circuit breaker of the pump is switched off.
	c) The motor protection of the pump has tripped due to overload.	Check if the motor/pump is blocked.
	d) The pump is defective.	Repair or replace the pump.
	e) The PM unit is defective.	Repair or replace the PM unit.
8. System without pressure tank: The red "Alarm" indicator light flashes once per period.	a) Cycling. A tap has not been closed entirely after use.	Check that all taps have been closed.
	b) Cycling. There is a minor leakage in the system.	Check the system for leakages.
9. System with pressure tank: The red "Alarm" indicator light flashes once per period.	a) Cycling. The pressure tank has no precharge pressure, or the tank size is insufficient.	Check the tank precharge pressure, and recharge the tank, if necessary. If the size of the pressure tank is insufficient, set DIP switch 5 to "OFF", or replace the pressure tank. See quick guide.
10. The red "Alarm" indicator light flashes twice per period.	a) Maximum continuous operating time (30 minutes). The pump has been running continuously for 30 minutes.	Check the system for leakages. Disable the function to allow the pump to run for 30 minutes. See quick guide.
11. The red "Alarm" indicator light flashes three times per period, and each pump start is delayed a few seconds.	a) Too many start/stop sequences within a short period. The pressure tank has no precharge pressure, or the tank size is insufficient.	Check the tank precharge pressure, and recharge the tank, if necessary. If the size of the pressure tank is insufficient, set DIP switch 5 to "OFF", or replace the pressure tank.
	b) Too many start/stop sequences within a short period. The PM 2 is set to start/stop with 1 bar differential pressure, i.e. DIP switch 5 is set to "ON", but no pressure tank has been installed in the system.	Set DIP switch 5 to "OFF".
12. The red "Alarm" indicator light flashes four times per period.	a) Pressure sensor fault.	Repair or replace the PM unit.
13. The pump performance is unstable.	a) Pump inlet pressure too low.	Check the inlet conditions of the pump.
	b) Suction pipe is partly blocked by impurities.	Remove and clean the suction pipe.
	c) Leakage in suction pipe.	Remove and repair the suction pipe.
	d) Air in suction pipe or pump.	Vent the suction pipe/pump. Check the inlet conditions of the pump.

Fault	Cause	Remedy
14. The pump runs backwards when switched off.	a) Leakage in suction pipe.	Remove and repair the suction pipe.
	b) Foot or non-return valve defective.	Remove and clean, repair or replace the valve.
	c) The foot valve is stuck in completely or partly open position.	Remove and clean, repair or replace the valve.
Self-priming pumps only:		
15. The pump performance is unstable and the pump is noisy.	a) The differential pressure across the pump is too low.	Close the tap gradually until the discharge pressure is stable and the noise has ceased.
16. When startup is attempted, the pump will start, but delivers no pressure or flow.	a) Liquid column above non-return valve in discharge pipe prevents the pump from self-priming.	Empty the discharge pipe. Make sure that the non-return valve does not hold back liquid in the discharge pipe. Repeat the startup procedure.
	b) Suction pipe draws in air.	Make sure that the suction pipe is airtight from pump to liquid level. Repeat the startup procedure.
17. The pump runs, but does not deliver the rated flow.	a) The internal valve did not close.	Close the tap gradually until the pressure or flow suddenly increases. Then open the tap gradually until the required flow is reached.

8. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Declaration of conformity

GB: EC/EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products CMB Booster PM2, CMB-SP Booster PM2, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EC/EU member states.

CN: 欧共体 / 欧盟符合性声明

我们，格兰富，在我们的全权责任下声明，产品 CMB Booster PM2, CMB-SP Booster PM2，即该合格证所指之产品，符合欧共体 / 欧盟使其成员国法律趋于一致的以下理事会指令。

ID: Deklarasi kesesuaian Komunitas Eropa/Uni Eropa

Kami, Grundfos, menyatakan dengan tanggung jawab kami sendiri bahwa produk CMB Booster PM2, CMB-SP Booster PM2, yang berkaitan dengan pernyataan ini, sesuai dengan Petunjuk Dewan berikut ini serta sedapat mungkin sesuai dengan hukum negara-negara anggota Komunitas Eropa/Uni Eropa.

- Machinery Directive (2006/42/EC).
Standards used: EN 809:1998 + A1:2009.
 - Low Voltage Directive (2014/35/EU)
Standards used: EN 60335-1:2012+ A11:2014, EN 60335-2-51:2003 + A1:2008 + A2:2012.
 - EMC Directive (2014/30/EU)
Standards used: EN60730-1:2000, EN 60730-1 + A16:2007.
- This EC declaration of conformity is only valid when published as part of the Grundfos Installation And Operating instructions (publication number 97501373 1216).

Bjerringbro, 21st January 2016



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